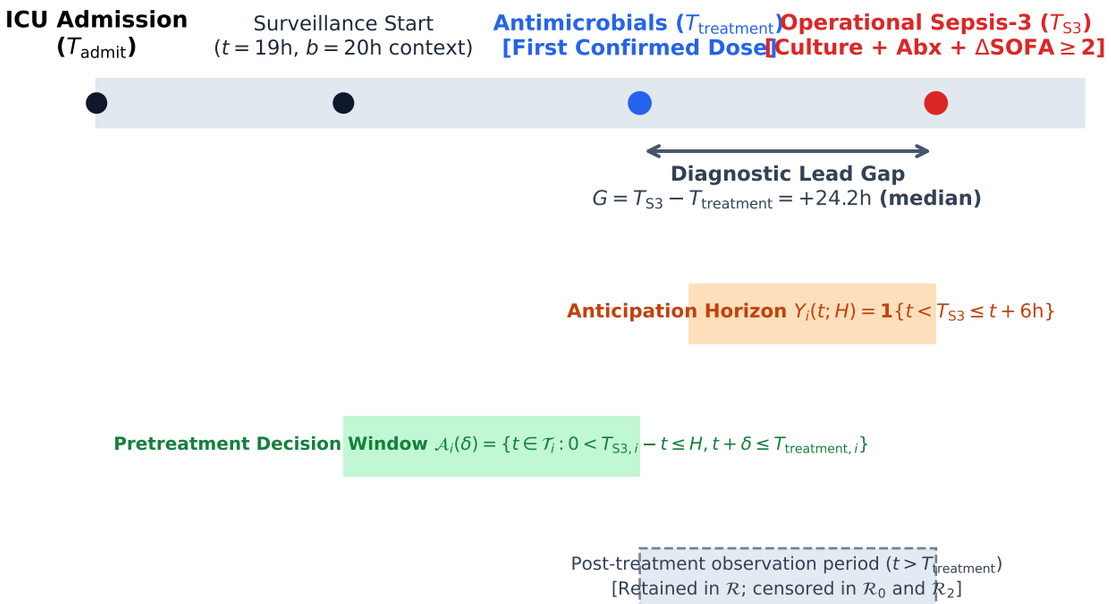
A. Operational Clocks & Actionable Decision Boundary



Mathematical Formalization (§2.3):

- Evaluated Risk Set: $\mathcal{R} = \{(i,t) : i \in \mathcal{C}, t \in \mathcal{T}_i, t < T_{S3,i}\}$ (97,700 windows across 8,254 stays)
- Anticipation Label: $Y_i(t;H) = \mathbf{1}\{t < T_{S3,i} \le t + H\}$ ($H = 6$h; 678 pos / 97,022 neg, $\pi = 0.694\%$)
- Pretreatment Sets: $\mathcal{R}_\delta = \{(i,t) \in \mathcal{R} : t + \delta \le T_{treatment,i}\}$ ($\mathcal{R}_0$: 70,192; $\mathcal{R}_2$: 70,129 windows)

B. Population Opportunity Attrition

